Green University of Bangladesh

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Project Proposal

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Lab Project Status	
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Title: A Lightweight Cross-Platform Virtual Classroom and Collaboration System for Students and Educators

1 Introduction

The rise of online learning and remote collaboration has highlighted the importance of effective digital platforms that enable communication, content sharing, and collaboration. Although platforms like Google Classroom and Microsoft Teams are widely adopted, both have significant shortcomings. Google Classroom lacks integrated video conferencing, while Microsoft Teams, despite being powerful, is often criticized as overly complex for educators and students.

This project proposes the development of a cross-platform virtual classroom system that covers these gaps. Built using Flutter for mobile, web, and desktop, the system will deliver a seamless, lightweight, and complete virtual classroom and collaboration experience.

2 Goal and Objectives

Goal

- To design and develop a comprehensive Virtual Classroom and Collaboration System that operates seamlessly across web, mobile, and desktop platforms.

Objectives

1. To implement true cross-platform accessibility and responsiveness using the Flutter framework.
2. To develop and integrate core communication features such as one-to-one and group messaging, emoji reactions, file sharing, and high-definition video conferencing.
3. To embed intelligent scheduling and calendar synchronization with widely used services like Outlook and Google Calendar.
4. To ensure robust user authentication, end-to-end data encryption, and compliance with international education data protection standards.

3 Problem Analysis and Motivation

Problem Analysis

In the modern digital learning landscape, several educational and collaboration platforms have emerged to facilitate communication between students and educators. However, despite their widespread adoption, many of these systems suffer from usability, accessibility,

and integration challenges that hinder an optimal learning experience. A comparative analysis of popular platforms reveals the following limitations:

- **Google Classroom:** While widely used for course management, it lacks built-in support for real-time interactive meetings. Users often need to depend on external tools such as Google Meet or Zoom, which leads to fragmented workflows and inconsistent user experiences.
- **Microsoft Teams:** Offers an extensive range of collaboration tools, including video conferencing and document sharing, but is frequently criticized for its complexity. Educators and students, especially in non-technical environments, face steep learning curves and interface overload.
- **Moodle:** An open-source learning management system that provides flexibility but often requires significant technical expertise to set up and maintain. Many users on mobile platforms report outdated UI designs and poor app responsiveness.
- **Edmodo:** Known for its social learning approach, but user reviews highlight slow performance, confusing navigation, and insufficient moderation tools for large classes.
- **General Issues across Platforms:** Many existing platforms confine users within closed ecosystems, require costly institutional licenses, and perform poorly in low-resource environments. Furthermore, compatibility issues across devices and operating systems limit cross-platform accessibility and collaboration.

These challenges emphasize the need for a lightweight, cross-platform, and user-friendly solution that combines real-time communication, content sharing, and interactive collaboration without dependency on multiple external tools. The motivation for this project arises from addressing these gaps through an integrated platform that prioritizes accessibility, simplicity, and scalability for both students and educators.

Motivation

Our motivation is to build a lightweight, cost-effective, and complete cross-platform collaboration platform that:

- Combines the strengths of Teams and Classroom while eliminating their weaknesses.
- Provides a user-friendly interface accessible to all learners and teachers.
- Promotes inclusivity by working efficiently even in low-bandwidth environments.
- Encourages open integrations and avoids vendor lock-in.

4 Literature Review

Recent research on virtual learning platforms offers valuable insights into their effectiveness, user experience, and pedagogical challenges. This section reviews two widely adopted systems Google Classroom and Microsoft Teams to identify their strengths and limitations as learning management and collaboration tools.

Google Classroom

Google Classroom has gained significant attention for its simplicity, accessibility, and efficiency in managing assignments and educational resources. According to Abdullah et al. [1], the platform positively influences teaching and learning outcomes by streamlining communication between educators and students. Amrizal and Pujiastuti [2] further observed that Google Classroom enhances English language learning, particularly in developing writing and speaking skills. Despite these advantages, challenges such as navigation complexity, limited customization options, and infrastructure-related barriers continue to hinder its widespread effectiveness [3].

Microsoft Teams

Microsoft Teams provides an integrated digital workspace that combines synchronous and asynchronous learning tools, including chat, video meetings, and collaborative file sharing. Evans [4] highlighted that Teams supports authentic assessment and interactive engagement in higher education contexts. However, other studies, such as Pal et al. [5], identified barriers related to user adaptation, technological familiarity, and sustained engagement in virtual learning environments.

Platform/Study	**Key Points**
Google Classroom (Abdullah et al., 2022)	Improves teacher-student communication and learning outcomes.
Google Classroom (Amrizal & Pujiastuti, 2022)	Enhances English writing and speaking skills.
Google Classroom (RSIS, 2021)	Navigation and customization issues; infrastructure barriers.
Microsoft Teams (Evans, 2022)	Supports interactive learning and authentic assessment.
Microsoft Teams (Pal et al., 2020)	Challenges in user adaptation and sustained engagement.

Table 1: Key Points from Literature on Virtual Learning Platforms

5 Feasibility Study

To ensure the successful execution and long-term viability of the proposed cross-platform virtual classroom system, a comprehensive feasibility analysis has been conducted across technical, economic, operational, and schedule dimensions.

- **Technical Feasibility:** The proposed system utilizes widely adopted, well-documented, and open-source technologies. Flutter enables single-codebase deployment across Android, iOS, web, and desktop, significantly reducing development overhead. Node.js/Go along with PostgreSQL and MongoDB provide a reliable and scalable backend architecture. Furthermore, integrating WebRTC with Selective Forwarding Units (such as Mediasoup or LiveKit) makes real-time audio and video streaming technically achievable with low latency even in low-bandwidth scenarios.

- **Economic Feasibility:** By leveraging open-source frameworks and tools, the project minimizes software licensing expenses. The total estimated budget of 380,000 BDT covers essential infrastructure, SDK licenses, development, and pilot deployment. Compared to proprietary institutional licenses required by existing enterprise solutions, this system offers a highly cost-effective alternative with a strong return on investment for educational institutions.

- **Operational Feasibility:** The primary motivation of this platform is to eliminate the steep learning curve of Microsoft Teams and the fragmented workflow of Google Classroom. By focusing on an intuitive, lightweight user interface and integrating familiar calendar services (Google Calendar and Outlook), both educators and students can adopt the system seamlessly without requiring extensive technical training.

- **Schedule Feasibility:** The project is structured across a realistic one-year roadmap divided into six distinct phases using the Agile Scrum methodology. Organizing development into two-week sprints ensures incremental delivery of the Core MVP, timely testing, and sufficient buffer for quality assurance prior to the final production launch.

6 Methodology

The methodology for this project will follow a structured and iterative approach to ensure quality, scalability, and timely delivery. The process combines Agile principles with rigorous testing and documentation to achieve a balance between flexibility and reliability.

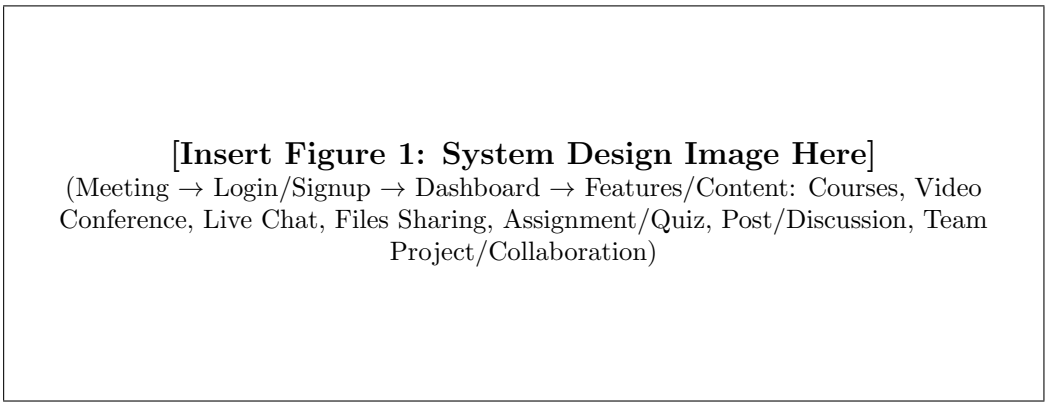


Figure 1: System Design of this project.

Development Approach

The project will adopt the Agile Scrum methodology, organized into two-week sprints. Each sprint will deliver incremental functionality, enabling continuous feedback, rapid adaptation to changes, and early detection of potential risks. Regular sprint reviews and retrospectives will be conducted to ensure alignment with project objectives and stakeholder expectations.

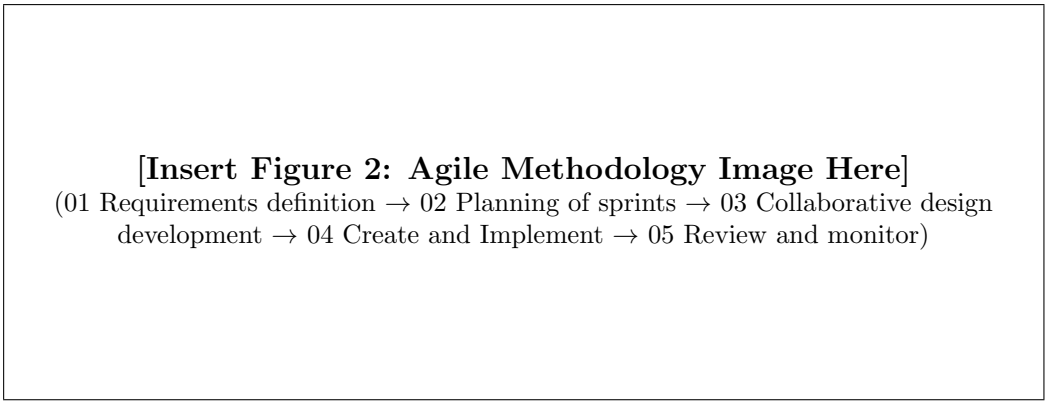


Figure 2: Agile Methodology.

Technology Stack

The system will be developed using a robust and modern technology stack to ensure cross-platform compatibility, performance, and security:

- **Frontend:** Flutter framework for Android, iOS, web, and desktop applications.
- **Backend:** Node.js or Go for REST APIs, supported by WebSockets for real-time communication.
- **Database:** PostgreSQL for structured data such as users, courses, schedules. MongoDB for unstructured or semi-structured data like chat messages and activity logs.
- **Media Services:** WebRTC with a Selective Forwarding Unit (SFU) such as Mediasoup or LiveKit for real-time audio and video conferencing.
- **Integrations:** Microsoft Graph API for seamless Office 365, OneDrive, and Outlook integration.
- **Security:** OAuth 2.0 for authentication and end-to-end encryption for communication and data protection.

Project Phases

The implementation will be divided into well-defined phases to ensure systematic progress:

1. **Requirement Analysis and Design:** Eliciting functional and non-functional requirements, creating system architecture, and preparing design documents.
2. **Frontend Development:** Building the user interface using Flutter, focusing on responsiveness and accessibility across platforms.
3. **Backend Development:** Implementing REST APIs, database schema, and WebSocket-based services for real-time features.
4. **Real-Time Communication Module:** Developing video conferencing, chat, and file-sharing functionalities using WebRTC.

5. **Integration with Office 365 and Calendar Services:** Enabling co-authoring, cloud storage, and synchronization with Outlook and Google Calendar.
6. **Collaboration Tools:** Implementing digital whiteboards, shared notes, and polling features to facilitate interactive learning.
7. **Testing and Quality Assurance:** Conducting unit tests, integration tests, usability testing, and security assessments.
8. **Deployment and Pilot Testing:** Deploying the system on cloud infrastructure (AWS/Azure/GCP) and conducting pilot runs with selected institutions.
9. **Documentation and Training:** Preparing user manuals, technical documentation, and providing training for educators and administrators.

7 Gantt Chart

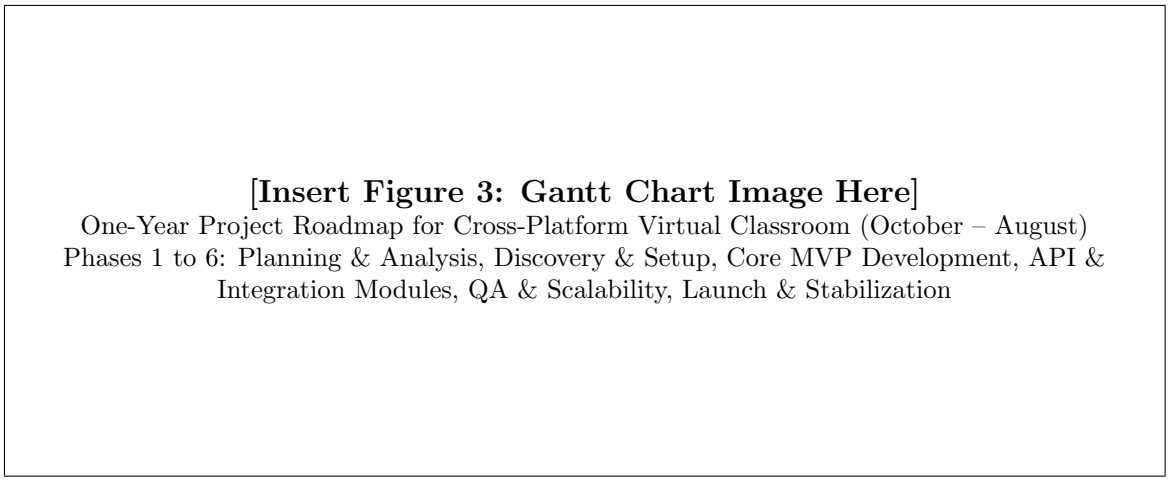


Figure 3: Gantt Chart representing project phases and timeline.

8 Budget Plan

Module	**Estimated Cost (BDT)**	**% Total**
System Setup & Infrastructure	50,000	13.16%
Backend & API Development	80,000	21.05%
Frontend & UI Development	40,000	10.53%
Integration Modules	10,000	2.63%
Video Conferencing Setup	60,000	15.79%
Tools & SDK Licenses	50,000	13.16%
Testing & QA	40,000	10.53%
Deployment & Maintenance	30,000	7.89%
Marketing & Pilot Launch	20,000	5.26%
Total	**380,000**	**100%**

Figure 4: Budget Plan showing estimated costs for development and deployment.

9 Conclusion

This project proposes a complete cross-platform virtual classroom and collaboration system that addresses the shortcomings of existing solutions like Microsoft Teams and Google Classroom. By combining simplicity, accessibility, and integration with powerful collaboration tools, our system will empower students and educators with a secure, scalable, and user-friendly platform. The outcome will significantly enhance the virtual learning experience, making education more inclusive and effective.

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